

Capstone SOC Incident Response Report

Executive Summary: On 2025-08-18 a simulated multi-stage intrusion targeted a lab environment running Metasploitable2. The adversary leveraged a Samba usermap exploit (T1210) to gain code execution, followed by valid account misuse (T1078) to move laterally. Wazuh SIEM detected anomalous Samba events and unusual account activity; alerts were correlated and escalated through TheHive. Immediate containment actions included isolating the impacted VM and blocking the attacker IP (192.168.1.102) via CrowdSec. Forensics data (netstat, process listings, and memory snapshot) were preserved for analysis. Timeline: - 14:00 UTC: Samba exploit attempt detected (Metasploit) - 14:05 UTC: Wazuh generated high-priority alert (samba events + process spawn) - 14:20 UTC: TheHive case created and escalated to Tier 2 - 14:30 UTC: CrowdSec blocked attacker IP; VM isolated Root Cause & RCA (summary): The root cause was an exposed, unpatched Samba service combined with insufficient network segmentation. Applying the 5 Whys highlighted gaps in patch management and firewall rules permitting lateral access. Recommendations: 1. Patch and harden Samba and public-facing services immediately. 2. Implement strict network segmentation and least-privilege access. 3. Enforce MFA for privileged accounts and rotate credentials. 4. Integrate threat intelligence feeds for automated enrichment and refine SIEM rules to reduce detection time. Prepared by: SOC Analyst